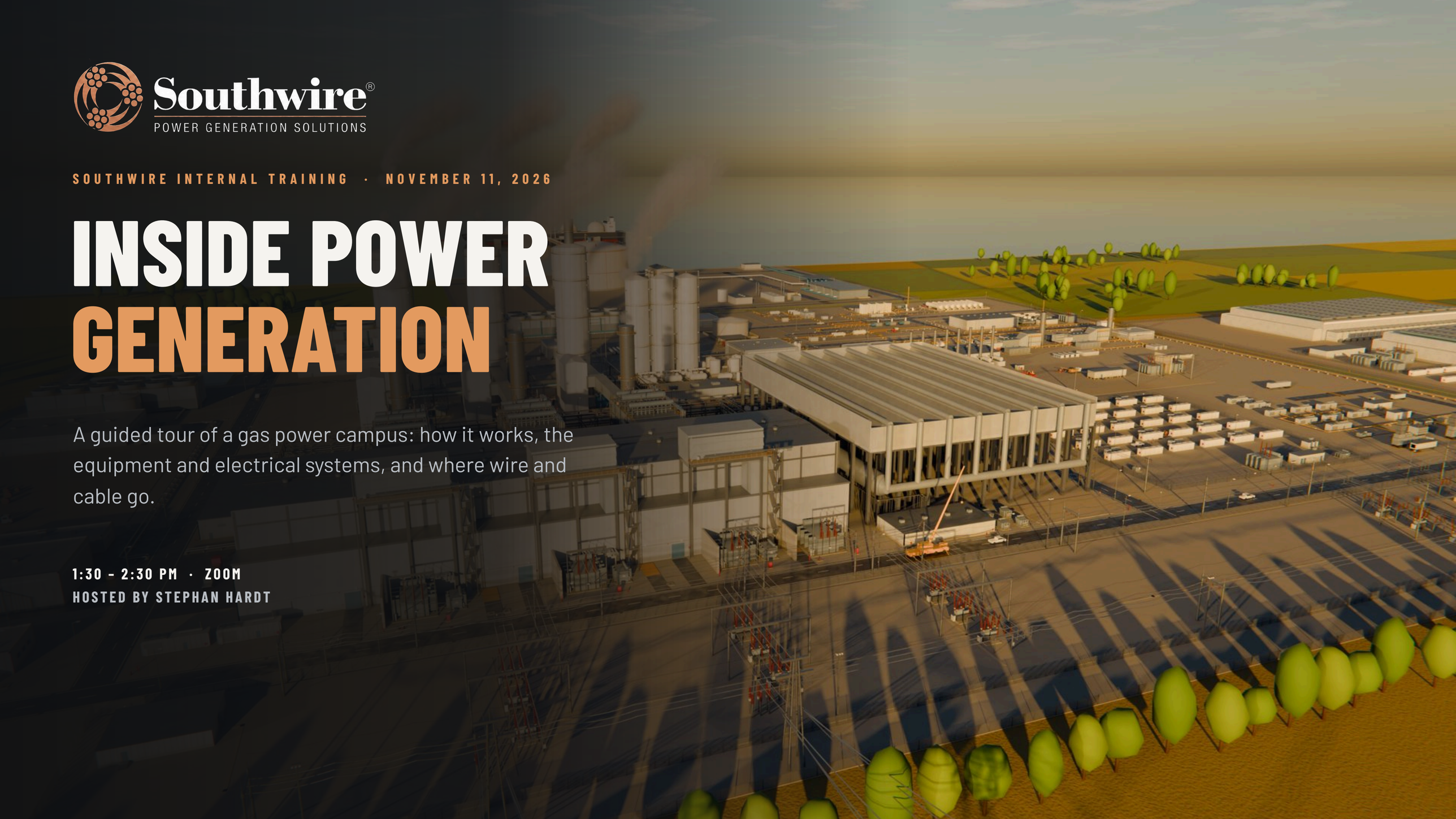
Southwire®
POWER GENERATION SOLUTIONS

SOUTHWIRE INTERNAL TRAINING · NOVEMBER 11, 2026

INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

1:30 - 2:30 PM · ZOOM
HOSTED BY STEPHAN HARDT



WHY BUILD A WHOLE PLANT? TO TELL THE CABLE STORY



1 NOTHING LIKE IT EXISTED

NO COMPETITOR MATERIAL, NO PUBLIC REFERENCE

OEM brochures stop at the turbine; EPC drawings are confidential. We found nothing that shows a whole gas plant with its cable, so we built it.



2 A WAY TO TALK WITH CUSTOMERS

ONE PICTURE EVERYONE CAN POINT AT

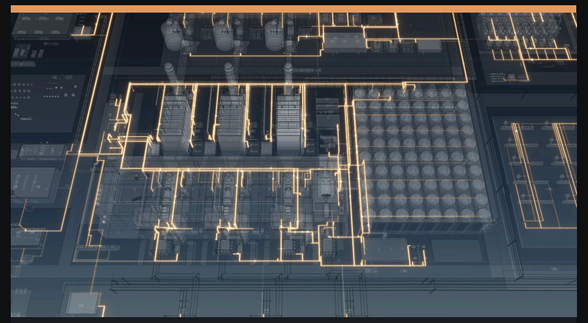
“This is your plant, this is where our cable goes.” Sales, an EPC and an owner looking at the same view, in their language.



3 A SEAT AT THE IPP TABLE

EARLY, BEFORE THE EPC BUYS

IPPs and developers plan campuses: power block, batteries, data centre, behind-the-meter. The model matches how they think, so we join the conversation early.



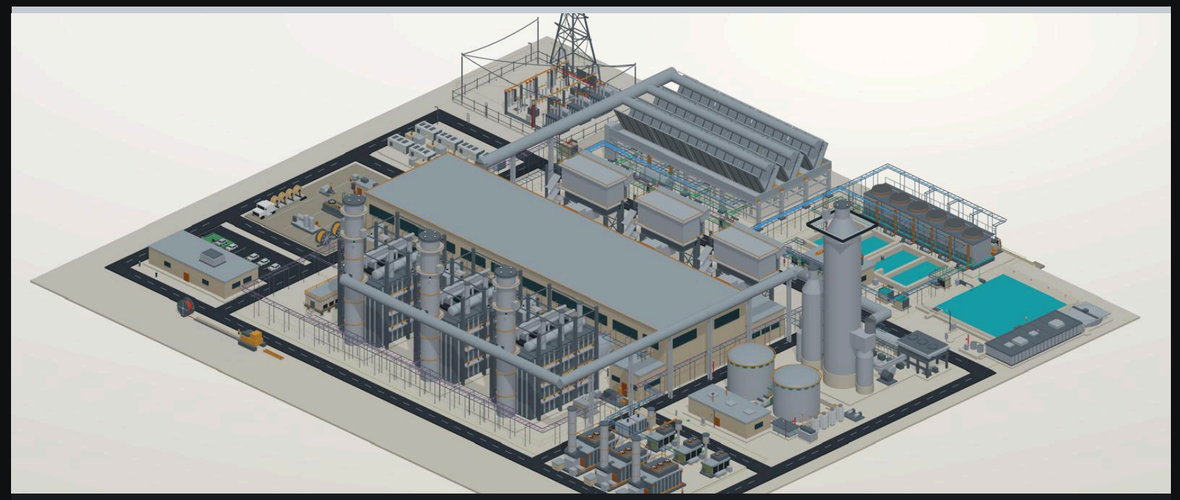
4 CABLE IS THE STORY

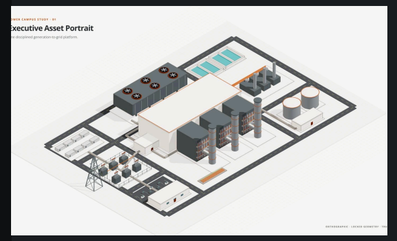
THE PLANT IS THE BACKDROP

Every render, brochure and slide shows our products applied where they go: MV-105 to the motors, TC-ER in the trays, ARMOR-X at the burners.

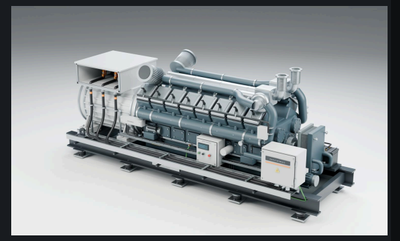
Not engineered for one customer: built so we can talk about theirs. One model behind the brochures, the decks, the 3D viewer and this training.

THEN A PLANT YOU CAN MEASURE







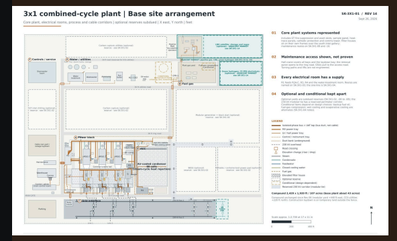


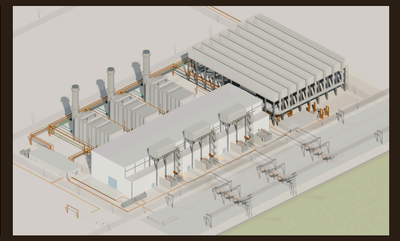
V2 3D STUDIES

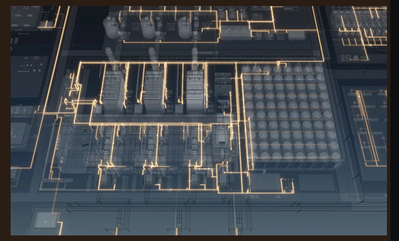
SPRING - SUMMER

Real plant elements in 3D: the hall, gas turbines, HRSGs, engines, the plant zones. Closer, but not to scale and not routed: still no cable to count.









V3 A PLANT YOU CAN MEASURE

SEPTEMBER - TODAY

Drafted first: the SK-3X1 drawing set (16 sheets, 230 / 13.8 / 4.16 kV, 480 V), then the model built from it. LOD 3, a 6,011-run cable schedule, the X-ray, the tour.

WHY WE BUILT IT: SEE THE WHOLE PLANT · MEASURE, DON'T GUESS · ONE SOURCE FOR BROCHURES, DECKS, VIEWER AND SCHEDULE · SPEAK THE BUYER'S LANGUAGE

A BOOM-SCALE GAS BUILD, AND CABLE IS ON THE CRITICAL PATH

THE MARKET

378 GW

US gas power in development; 52 GW already under construction

THE CABLE

~$1.2B

core wire & cable, 2026–2035, at $8–13M per GW

THE PIPELINE

386

gas projects we track in 40 states, every owner type

THE TRACTION

$50M+

booked in Q3; $30M in September alone

TODAY'S GOAL: KNOW THE PLANT, SO YOU CAN FOLLOW THE CABLE THROUGH IT

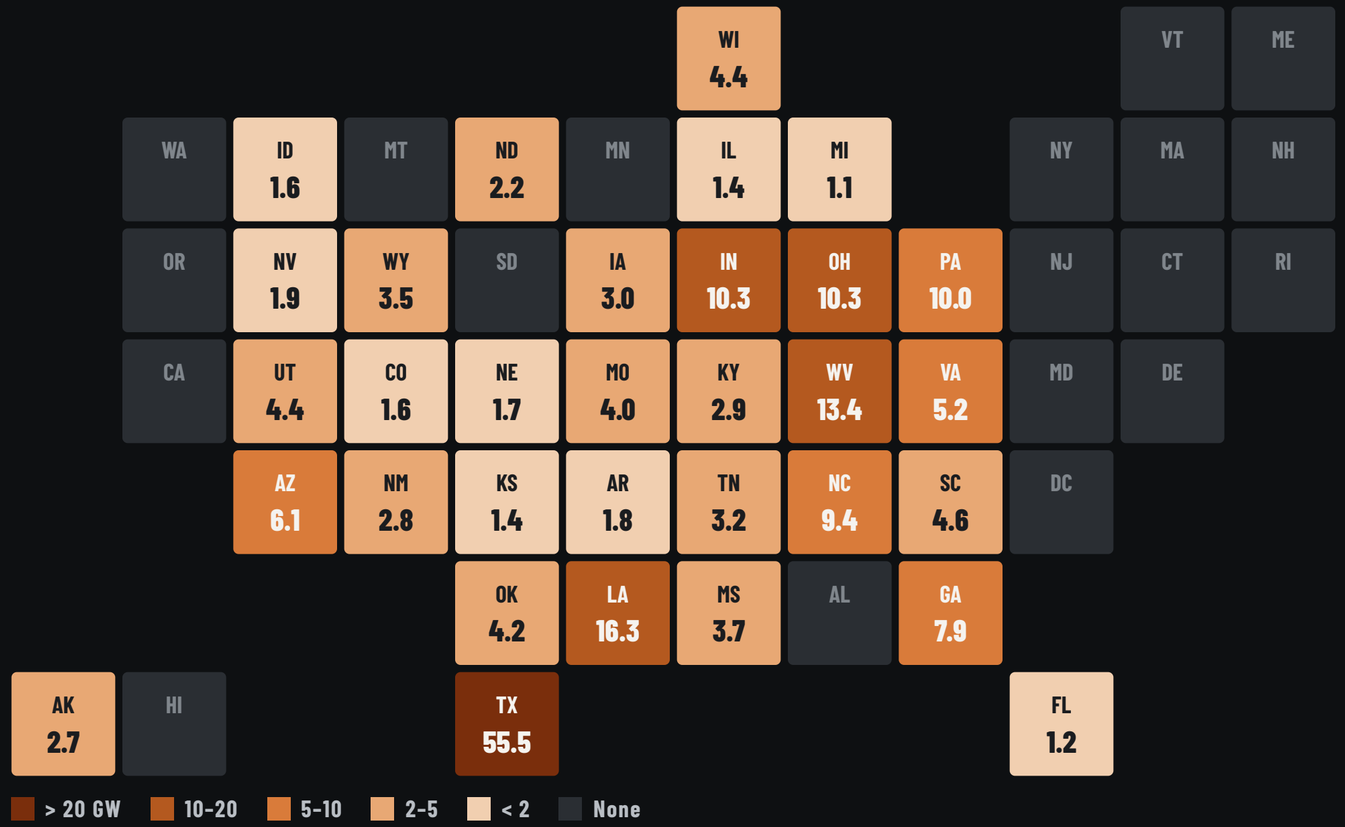
Global Energy Monitor, Aug 2026; Southwire PowerGen planning model and internal figures, 2026.

stephan.hardt@southwire.com

PROJECT CONCENTRATION

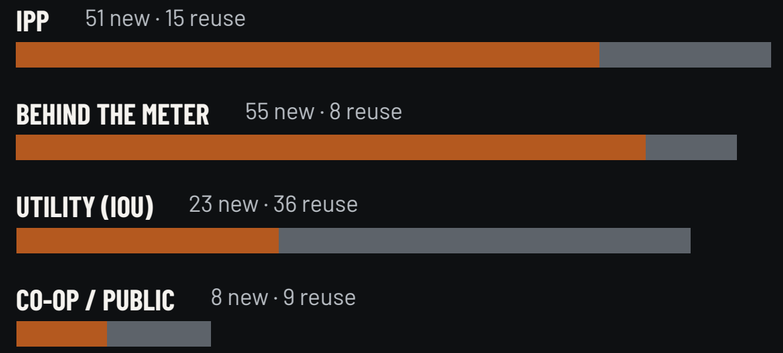
WHERE THE BUILD IS, WHO OWNS IT, WHAT KIND OF SITE

GROSS GW BY STATE · 386 PROJECTS, COD 2026-2035



OWNER × SITE · GROSS GW

GREENFIELD 137 · BROWNFIELD 69



UTILITIES REUSE OLD SITES; IPPS AND BTM BUILD NEW

Greenfield, grid-connected	77.0
Data-centre / BTM campus	54.5
Coal-site repower	41.3
Expansion at an operating plant	21.4
Industrial captive and site reuse	11.6

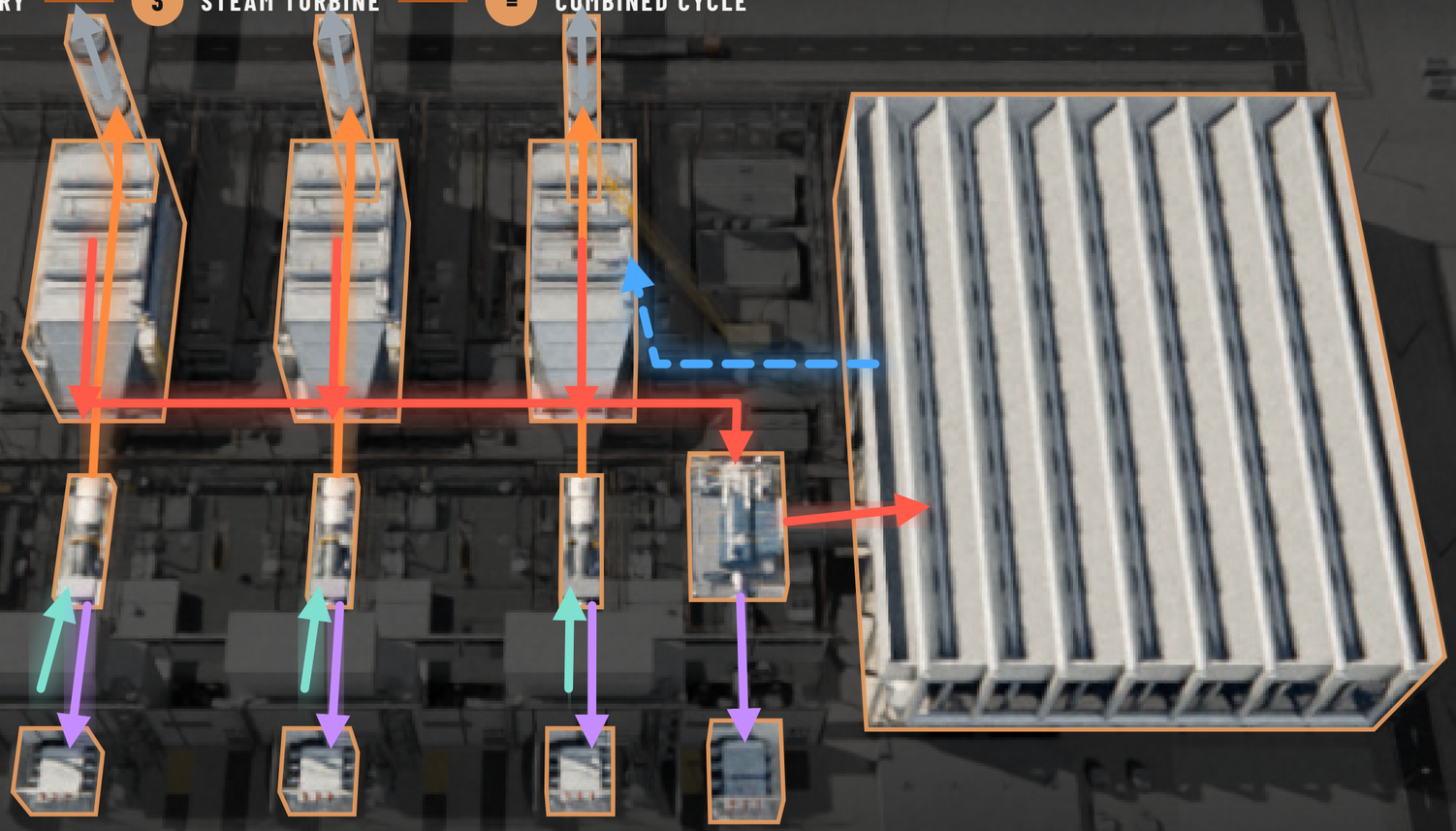
Southwire PowerGen register, Sep 11, 2026. Gross GW, unweighted. Site classes provisional (Rev01).

COMBINED CYCLE

BURN THE GAS ONCE, USE THE HEAT TWICE

EXHAUST · STEAM · WATER · POWER

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE — = COMBINED CYCLE



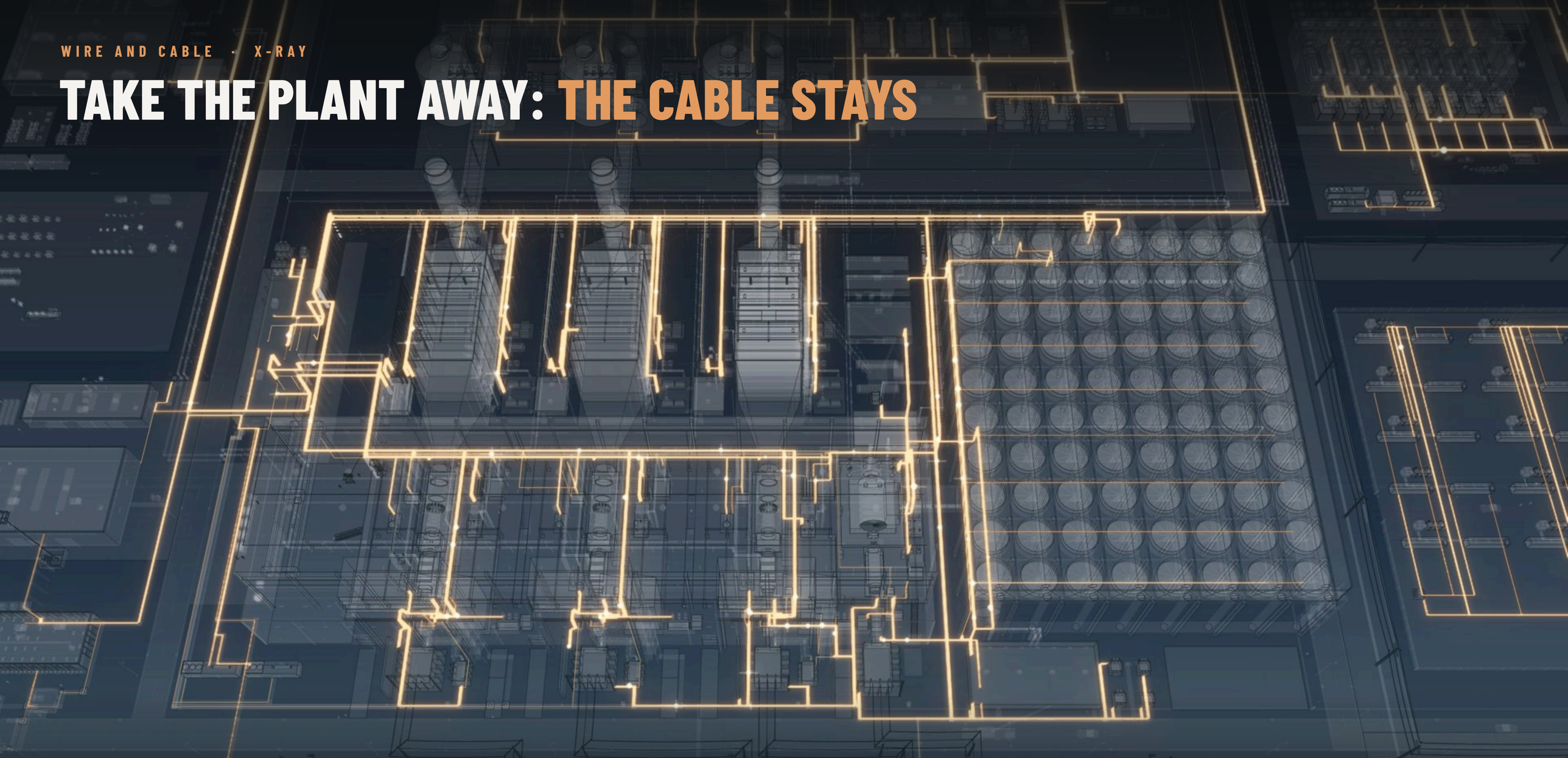
~60%
of the fuel becomes power, vs ~40% for a gas turbine alone

3 + 1
three gas turbines, three HRSGs, one steam turbine: a "3x1"

8,234
cables in this power block and its condenser, in our model

stephan.hardt@southwire.com

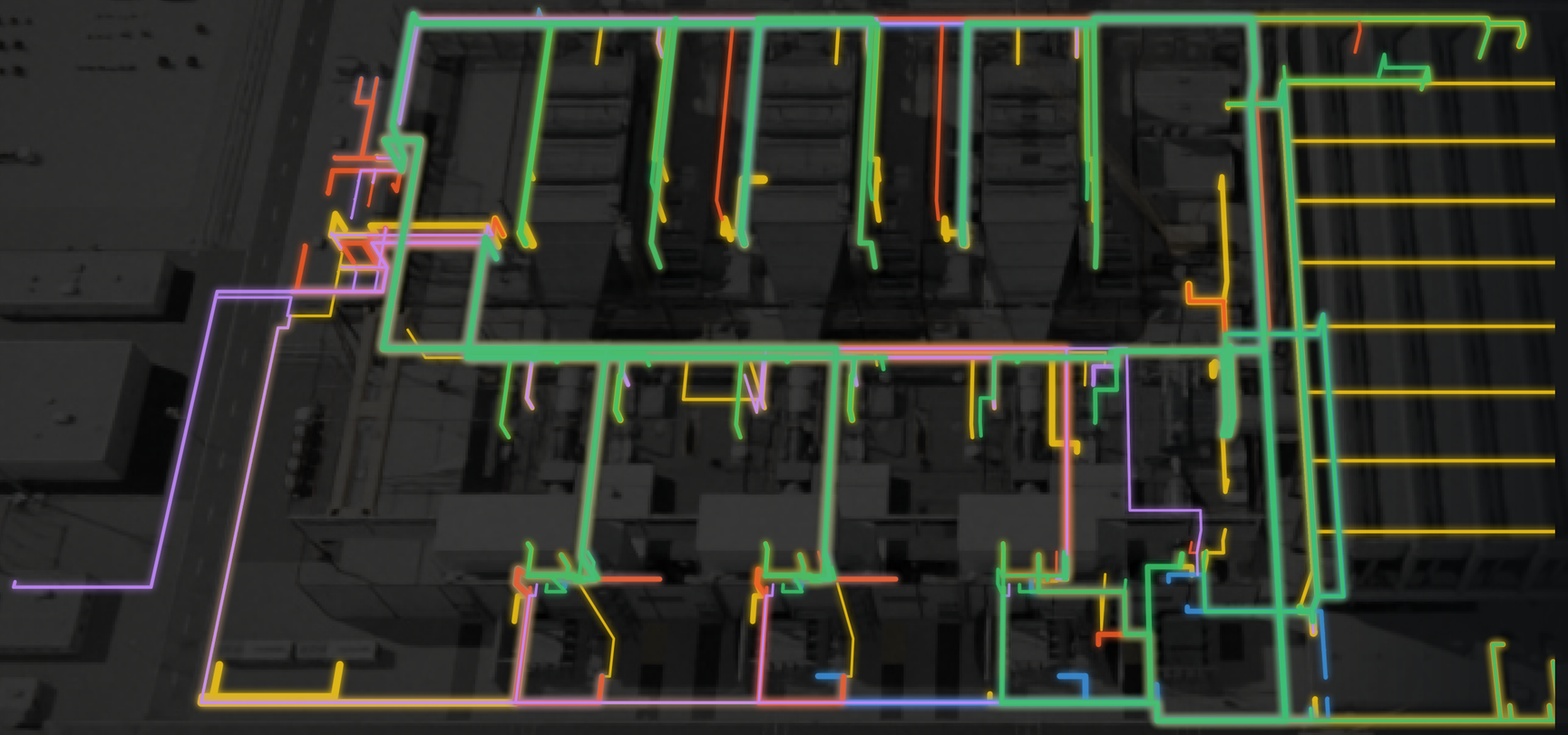
TAKE THE PLANT AWAY: THE CABLE STAYS



The same view, with the steel made glass. **About 8,200 cables and 440 miles** in the power block and its condenser, every one routed in our model.

stephan.hardt@southwire.com

NOW SORT IT BY JOB: FIVE KINDS OF CABLE



Every line is a route from the model's cable schedule, in trays, conduit and duct banks; thicker lines carry more cables. **The power block is a web of copper.**

THE CABLE IN THIS PICTURE

8,234 cables · ~439 miles

MV / HV POWER — 232 · 28 mi
The big feeders: generators, transformers, the largest motors. MV-105, HV XLPE.

LV POWER — 1,160 · 95 mi
600 V to motors, valves, drives, lighting. TC-ER, ARMOR-X MC-HL, VFD cable.

CONTROL — 1,255 · 94 mi
Starts, stops and trips: breakers, interlocks, MCC starters. Multi-conductor TC-ER.

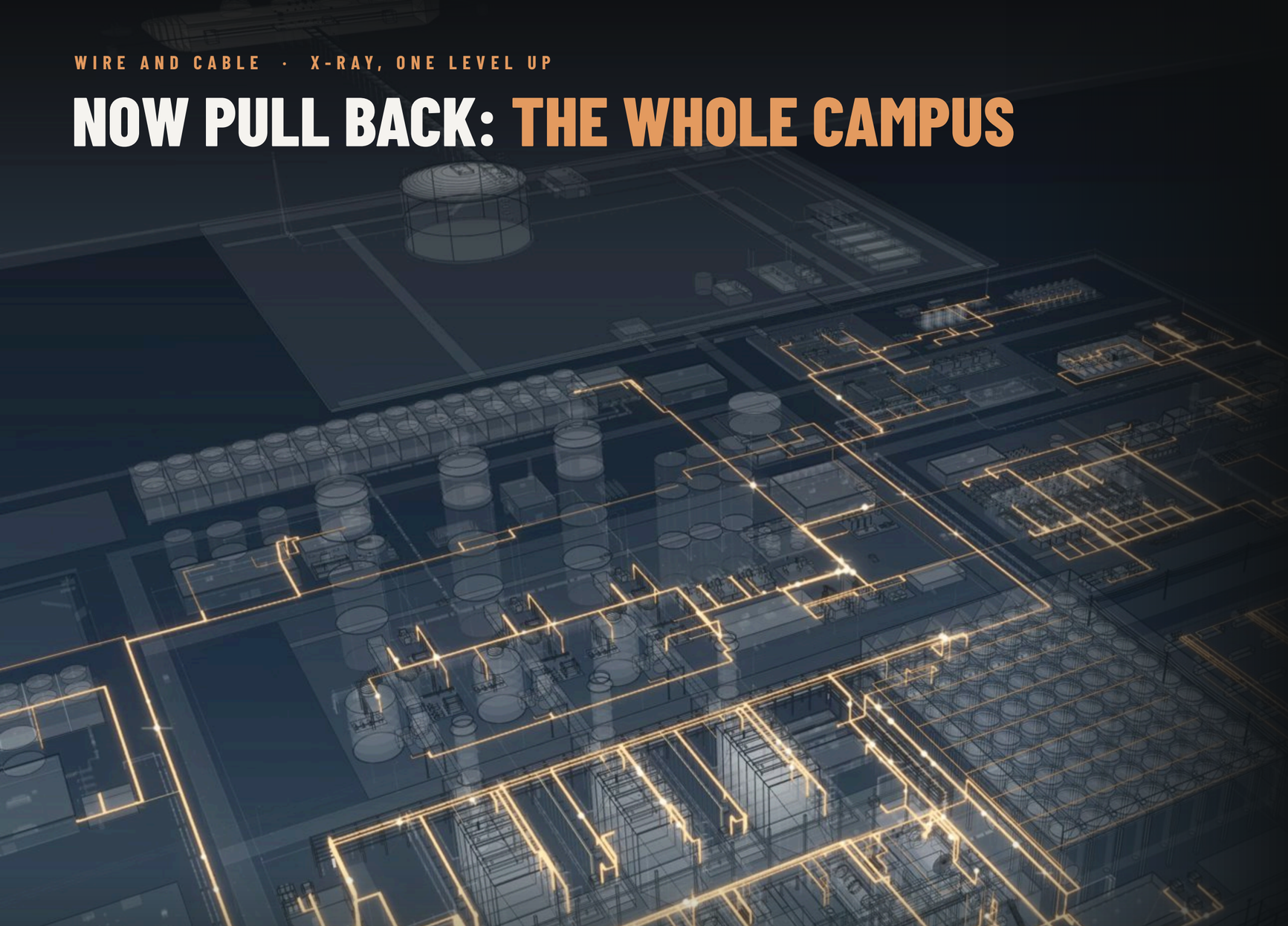
INSTRUMENTATION — 5,484 · 202 mi
The plant's nerves: pressure, flow, temperature. Shielded pairs, thermocouple extension.

FIBRE AND FIRE ALARM — 103 · 21 mi
The control network, protection and life safety.

The power block and its air-cooled condenser only, from the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 8/15

NOW PULL BACK: THE WHOLE CAMPUS



The power block and its condenser were the densest piece, **8,234 cables**. Add the grid, carbon capture, the batteries, the modular yard and the utilities, and **it is 2.6 times as much.**

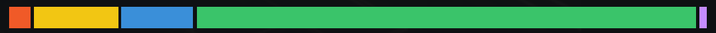
THE CABLE ON THE WHOLE CAMPUS

21,138 cables · ~1,143 miles

BY AREA ■ LAST SLIDE

POWER BLOCK	7,353
MODULAR / PORTABLE POWER	5,001
CARBON CAPTURE	2,753
FUEL SUPPLY	1,275
GRID INTERFACE	1,215
BESS	1,155
ACC	881
WATER / UTILITIES	575
GT INLET CHILLING	443
FUEL GAS	289
CONTROLS / SERVICE	198

BY TYPE

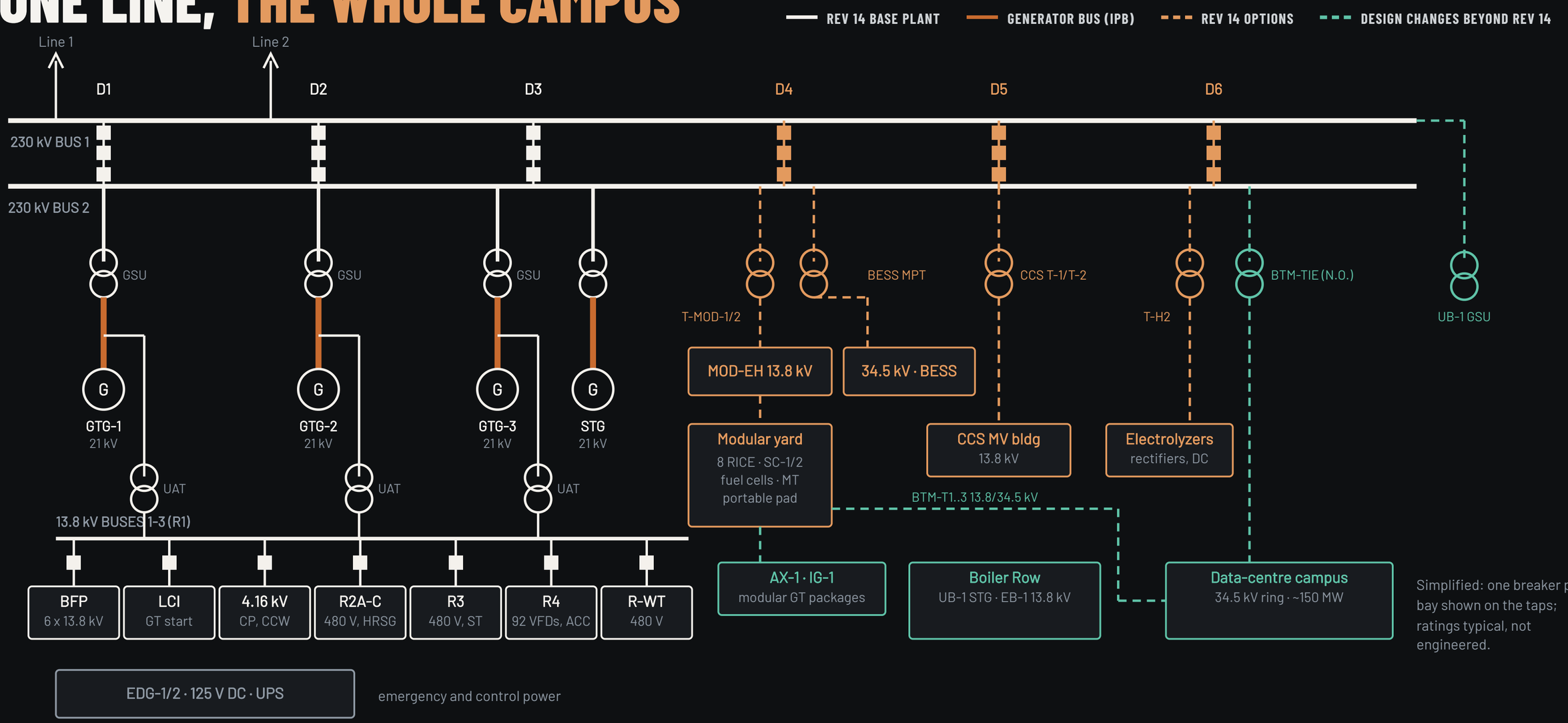


MV / HV power 3% · LV power 12% · Control 11% · Instrumentation 73% · Fibre and fire alarm 1%

From the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

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ONE LINE, THE WHOLE CAMPUS



LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

LEVEL	EQUIPMENT · RATING	FEEDS · CURRENT	CABLE IT DRIVES
230 kV	**GSU-1, 21 / 230 kV step-up**	Plant to the grid, overhead line	**Overhead conductor (outside the HRSG scope)**
21 kV	**GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA**	UAT-1 secondary 1,674 A into SWGR-13.8-1	**4 sets of 3 x 1/C 750 kcmil MV-105**
13.8 kV	**SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant**	Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A	**3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A**
480 V	**T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs**	Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles	**10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV**
120 V AC / 125 V DC	**UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels**	DCS, F&G node, breaker trip and close, burner-management power	**1/C #6 and control TC-ER 14 AWG**
Signals	**4-20 mA / HART, thermocouple millivolts, 24 V DC discretes**	Drum level, pressures, flame scanners, gas-path and tube-metal temperatures	**439 instrument + 84 thermocouple-extension cables, shielded pairs**

750 kW
connected at 480 V on one HRSG train, 78 circuits

2 x 5 MW
feed pumps: the largest motors on the train, at 13.8 kV

3,007 A
on the T-R2A secondary: 40 cables of 500 kcmil

74 %
of the train's cables are instrument or thermocouple pairs (68 % of the footage)

8.4 / 24.6
cal/cm^2 arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.

stephan.hardt@southwire.com · INSIDE POWER GENERATION 11/15

THREE BUCKETS OF CABLE, THREE DIFFERENT BUYERS



1 INSIDE THE PACKAGES

WHO BUYS

OEMS, E-HOUSE AND PACKAGE BUILDERS

E-house internals (switchgear and MCC interconnections, control and panel wire, short ties), turbine skids, modular gensets

WHEN

Every month, across many projects, months before site

OUR PRODUCTS

MV-105 ties, TC-ER, control and panel wire, pre-terminated harness kits

NOT IN OUR 440 MILES: A SEPARATE MARKET ON TOP



2 BETWEEN THE PACKAGES

WHO BUYS

THE EPC OR ITS ELECTRICAL CONTRACTOR

Field cable from the e-houses out to every motor, valve, instrument and transformer: the X-ray

WHEN

Once per plant, when detailed design fixes the cable schedule

OUR PRODUCTS

MV-105, TC-ER, ARMOR-X MC-HL, VFD, PLTC / ITC, thermocouple; SIMpull, cut-to-length

MOST OF THE MILES: ~440 IN THE POWER BLOCK



3 TO THE GRID

WHO BUYS

THE EPC'S HV CONTRACTOR OR THE UTILITY

GSU to switchyard, the gen-tie to the POI, grounding

WHEN

Early: HV cable has the longest lead time

OUR PRODUCTS

HV XLPE (69-345 kV), bare copper grounding

FEW CABLES, HIGH VALUE

Starting in the e-house is not the same as being bought by the e-house builder. Cover both: EPCs for the field cable; package and e-house builders for the cable inside. The builders buy every month and shape the spec at the terminations.



THE CAMPUS · FROM THE SEA TO THE GRID

ONE SITE, SEVEN STOPS

1. GRID CONNECTION AND SWITCHYARD
2. TURBINE HALL
3. HRSG AND ITS E-HOUSE
4. COOLING AND WATER
5. MODULAR GENERATION
6. BEHIND THE METER: THE DATA CENTRE
7. FUEL: GAS YARD AND LNG

LNG arrives by ship at the marine terminal; the power leaves on 230 kV lines. Everything in between runs on cable.

ONE SUPPLIER, GRID TO THERMOCOUPLE

 **GRID CONNECTION**	HV XLPE transmission cable (230 kV) · bare copper grounding
 **POWER ISLAND: MV**	MV-105 EPR 133% (5–35 kV) · ARMOR-X MC-HL 15 kV
 **POWER ISLAND: LV AND DRIVES**	Type TC-ER power · ARMOR-X MC-HL 600 V · VFD-rated cable
 **CONTROL AND INSTRUMENTATION**	TC-ER control · PLTC / ITC shielded pairs · thermocouple extension
 **HAZARDOUS AREAS**	ARMOR-X MC-HL, Class I Div 2: gas skids, fuel yards, burner fronts
 **BATTERY STORAGE**	PV / RHW-2 DC conductor (2 kV) · MV-105 collection (35 kV)
 **MODULAR AND TEMPORARY POWER**	Type SHD-GC portable MV · Type W · cam-lock sets
INSTALLATION	SIMpull reels and SIMpull truck · cut-to-length · pre-terminated harness kits

Plus the services: Spec support: sizing, derating, spec reviews · Cut-to-length against the cable schedule · Harness kits, pre-terminated and labelled · Project logistics: reel tracking, sequenced deliveries · Field support: pulling calcs, terminations, tests.

Typical Southwire families for each application; confirm constructions, sizes and listings against current Southwire specifications.

stephan.hardt@southwire.com

AFTER TODAY

RESOURCES AND WHO TO CALL

RESOURCES

THE 3D PLANT MODEL

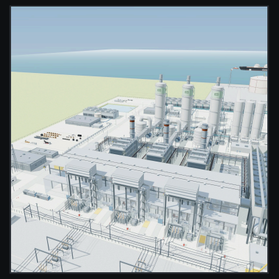
The interactive viewer you saw today, with the Nov 11 tour and the cable X-ray. Link sent with the recording.

THE BROCHURES

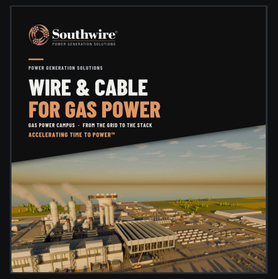
Wire & cable for gas power · EPC solutions · boiler solutions.

THE RECORDING AND THESE SLIDES

Shared after the session, with a glossary of the terms used today.



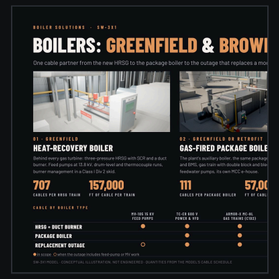
3D VIEWER



WIRE & CABLE BROCHURE



EPC SOLUTIONS



BOILER SOLUTIONS

POWER GENERATION SOLUTIONS TEAM

STEPHAN HARDT	PowerGen strategy, the plant model, owners and OEMs
PATRICK RITCHIE	EPC accounts
GREYSON	EPC outreach to more gas-plant builders
PHIL METZ · BOB BENISH	OEMs and package builders: e-houses, switchgear, modular power

stephan.hardt@southwire.com